PHP Control Structures

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Control Structures



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Conditional Processing

- Conditionals allow programmers to output different code based on specific conditions.

Types of conditionals:

1. if statement
2. if … else statement
3. if … elseif … else statement
4. switch statement

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Conditionals

if statement

- to execute some code only if a specified condition is true.

Syntax:

if (condition)
code to be executed if condition is true;

Sample Code:

if ($x == 1)
echo "True";

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Conditionals

if...else statement

- to execute some code if a condition is true and another code if the condition is false

Syntax:

```
if (condition)
        code to be executed if condition is true;
else
        code to be executed if condition is false;
```

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Conditionals

if...else statement

Sample Code:

```
if($x == 1)
        echo “True”;
else
        echo “False”;
```

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Conditionals

if...elseif....else statement

- to select one of several blocks of code to be executed

Syntax:

```
if (condition)
    code to be executed if condition is true;
    elseif (condition)
    code to be executed if condition is true;
    else
    code to be executed if condition is false;
```

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Conditionals

if...elseif....else statement

Sample Code:

```
if($x == 1)
        echo “One”;
elseif($x == 2)
        echo “Two”;
else
        echo “Three”;
```

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Conditionals

Short form of if – else statement using ternary operator

– Syntax for ternary operator

(condition ? Value_if_true : Value_if_false)

```
<?php
        $n = 15;
        $a = ($n % 2 ==0 ? 'Even Number': 'Odd Number')
?>
```

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Conditionals

switch statement

- to select one of several blocks of code to be executed

Syntax:

```
switch (n)
{
        case label1:
            code to be executed if n=label1;
            break;
        case label2:
            code to be executed if n=label2;
            break;
        default:
            code to be executed if n is different from both label1 and label2;
}
code to be executed if condition is false;
```

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Conditionals

switch statement

Sample Code:

```
switch($x)
{
        case 1:
           echo “One”;
           break;
        case 2:
           echo “Two”;
           break;
        default:
           echo “Three”;
           break;
}
```

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Conditionals

```
 $a = rand(1,6);
switch ($a){
   case 1:
      echo "<h$a>HEADER 1</h$a>";
      break;
   case 2;
      echo "<h$a>HEADER 2</h$a>";
      break;
   case 3;
      echo "<h$a>HEADER 3</h$a>";
      break;
   case 4;
      echo "<h$a>HEADER 4</h$a>";
      break;
   case 5;
      echo "<h$a>HEADER 5</h$a>";
      break;
   default;
      echo "OUT OF BOUNDS..";
      break;
}
```

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guess_num.php

```
<html>
    <head><title> Use of Decision Statements in PHP </title></head>
    <body bgcolor=cccc99>
        <h2> Guess the Number Result </h2>
        <?php
        $x =5; //guess
        $num = rand(1,10); //or you can set the value for the number,
        if ($x > $num) {
                echo "Your guess is too high. <br> It's $num. ";
        } elseif ($x < $num) {
                echo "Your guess is too low. <br> It's $num. ";
        } else {
                echo "You're Right! <br> It's $num. ";
        }
        ?>
    </body>
</html>
```

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Switch Statements



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number.php

```
<html>
    <?php
        $cellno = '09174685935';
        $prefix = substr($cellno,1,3);
        switch($prefix)
        {
            case "920":
            case "918":
                $telco = "Smart";
                break;
            case "917":
            case "915":
                $telco = "Globe";
                break;
            case "922":
            case "923":
                $telco = "Sun";
                break;
            default:
                $telco = "Unknown";
                break;

        }
        echo "$telco subscriber";
    ?>
</html>
```

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Loops

- Loops execute a block of code a specified number of times, or while a specified condition is true.

Types of loops:

1. while
2. do … while
3. for

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Loops

while loop

- Executes a block of code while a condition is true.

Syntax:

```
while (condition)
    {
    code to be executed;
    }
```

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Loops

While loop

Sample code:

```
$x = 1;
while ($x < 5)
{
        $x = $x + 1;
}
echo $x;
```

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Example: while loop

Quantity	Price
10	50
20	100
30	150
40	200
50	250
60	300
70	350
80	400
90	450
100	500

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prices_while.php

```
$brush_price = 5;
$counter = 10;

echo "<table border=\"1\" align=\"center\">";
echo "<tr><th>Quantity</th>";
echo "<th>Price</th></tr>";
while ( $counter <= 100 ) {
echo "<tr><td>";
echo $counter;
echo "</td><td>";
echo $brush_price * $counter;
echo "</td></tr>";
$counter = $counter + 10;
}
echo "</table>";
```

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Loops

do…while loops

- Always execute the block of code once, it will then check the condition, and repeat the loop while the condition is true.

Syntax:

```
do
    {
    code to be executed;
    }
    while (condition);
```

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Loops

do…while loops

Sample code:

```
$x = 1;
do
{
        $x = $x + 1;
}
while ($x < 5)
echo $x;
```

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prices_dowhile.php

```
$brush_price = 5;
$counter = 10;

echo "<table border=\"1\" align=\"center\">";
echo "<tr><th>Quantity</th>";
echo "<th>Price</th></tr>";
do {
echo "<tr><td>";
echo $counter;
echo "</td><td>";
echo $brush_price * $counter;
echo "</td></tr>";
$counter = $counter + 10;
} while ( $counter <= 100 );
echo "</table>";
```

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Loops

for loops

- used when you know in advance how many times the script should run. .

Syntax:

```
for (init; condition; increment)
    {
    code to be executed;
    }
```

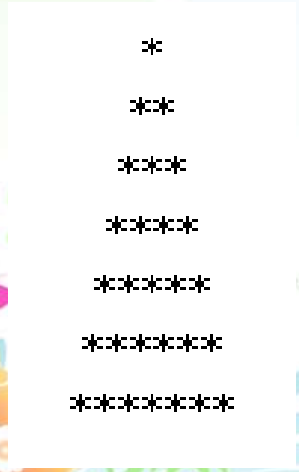
Sample code:

```
for($x = 1; $x < 5; $x++)
{
    echo $x . “<br />” ;
}
```

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Example: for loop

With the * symbol and the PRINT command any tower can be built. Write a PHP code that generates a tower with THREE or less print commands?



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Example: for loop

tower.php

```
<center>
<?php
$height = 7;
for ($i = 1; $i <= $height; $i++)
{
    for ($j = 1; $j <= $i; $j++)
    {
        print "*";
    }
    print "<br>";
}
?>
</center>
```

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Break and Continue

break breaks completely out of a loop and continues with the script statements after the loop

To break:

```
for ($a=1; $a < 6; $a++)
{
echo $a;
        if ($a > 3)
{
break;
}
}
```

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Break and Continue

continue skips to the end of the loop where the condition is tested. If the condition tests positive, the script continues from the top of the loop

To continue:

```
for ($a=1; $a < 6; $a++)
{
    if ($a == 3)
    {
    continue;
    }
    echo $a;
}
```

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